Disclaimer/credit. These questions have largely been ~~stolen~~ borrowed from natural sciences maths and vectors and matrices courses.

1) Prove the following results:

   (i) The trace of the product of a symmetric and an antisymmetric matrix is zero.

   (ii) If A is antisymmetric, then $\mathbf{x}^{\mathrm{T}}\mathrm{A}\mathbf{x} = 0$ for any column vector $\mathbf{x}$.

2) Two matrices A and B and a vector $\mathbf{x}$ have elements a_{ij}, b_{ij} and x_i respectively. (Note: this is not the same convention used in the lecture notes, it is important that you become familiar with different notations.)

   In this notation, what are the elements of A^{T}?

   In suffix notation, $\sum_k a_{ik}b_{kj}$ is an element of the matrix AB. Write down the matrices (or vectors) whose elements are

   (i) $\sum_{j,k} a_{ij}b_{jk}x_k$,

   (ii) $\sum_j x_j b_{ij}$,

   (iii) $\sum_{i,j} a_{ij}b_{kj}x_i$,

   (iv) $\sum_{j,k} a_{ij}a_{kj}a_{km}$.

3) Produce specific examples to show that

   (i) The product of two non-zero matrices can be zero;

   (ii) AB can be zero even if BA is non-zero;

   (iii) The product of two non-zero symmetric matrices can be anti-symmetric.

4) For any square matrix A, the matrix exp(A) is defined by

$$\exp(\mathrm{A}) = \mathrm{I} + \sum_{n=1}^{\infty} \frac{\mathrm{A}^n}{n!}.$$

   For the matrix

$$\mathrm{M} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix},$$

   and any real number θ, show that

$$\exp(\theta\mathrm{M}) = \mathrm{I} + \mathrm{M}\sin\theta + \mathrm{M}^2(1 - \cos\theta).$$

   Hence show that

$$\exp(\theta_1\mathrm{M})\exp(\theta_2\mathrm{M}) = \exp\big((\theta_1 + \theta_2)\mathrm{M}\big),$$

   and that $\big(\exp(\theta\mathrm{M})\big)\big(\exp(\theta\mathrm{M})\big)^{\mathrm{T}} = \mathrm{I}$.

   **For discussion with supervisors*: You have derived these results for the specific matrix M. Some of them are true more generally, can you see which?

5) For $\mathrm{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\mathrm{B} = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$, compute: $\det\mathrm{A}$, $\det\mathrm{B}$, $\det\mathrm{AB}$, and $\det(\mathrm{A}^{-1})$.

   Verify that $\det\mathrm{AB} = \det\mathrm{A}\det\mathrm{B}$ and $\det\mathrm{A}^{-1} = 1/\det\mathrm{A}$.

6) Calculate the determinant of

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

as $A_{11}C_{1,1} + A_{12}C_{1,2} + A_{13}C_{1,3}$ where $C_{i,j}$ is the relevant cofactor.

Compute the same determinant by using row operations to convert the determinant to a form with 0's below the diagonal. (See Example 3 in the notes).

Verify that $\det(\mathrm{A}^{\mathrm{T}}) = \det \mathrm{A}$.

7) What can be said about the columns of a matrix whose determinant is zero? For which value of a are the vectors $(1, 0, 1, 0)$, $(2, 1, 3, 2)$, $(4, 0, 1, 3)$ and $(2, 0, 3, a)$ linearly dependent?

8) In the theory of superconductivity, the energy E satisfies an equation of the form

$$\begin{vmatrix} a-E & -b & -b & -b & -b \\ -b & a-E & -b & -b & -b \\ -b & -b & a-E & -b & -b \\ -b & -b & -b & a-E & -b \\ -b & -b & -b & -b & a-E \end{vmatrix} = 0.$$

Show by solving this equation that there is one state with energy $E = a - 4b$ and all the others have energy $E = a + b$.

(Hint: it is not recommended to compute the determinant directly from its definition.)

9) Write the following equations in the form $\mathrm{A}\mathbf{x} = \mathbf{y}$:

$$\begin{aligned} x + y + az &= 1 \\ 3x + 4y + (2 + 3a)z &= 5 \\ -x + y + z &= b \end{aligned}$$

Calculate $|\mathrm{A}|$. Under what circumstances is $|\mathrm{A}| = 0$? For which values of a, b do the equations have a unique solution?

For the cases where a unique solution does not exist, determine the values of b for which there are no solutions. Hence work out the values for which there are many solutions.

10) Find the eigenvalues and eigenvectors of the matrix

$$\mathrm{M} = \begin{pmatrix} 5 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

Verify that: (i) the eigenvalues are real; (ii) the sum of the eigenvalues is equal to $\operatorname{trace} \mathrm{M}$; (iii) the product of the eigenvalues is equal to $\det \mathrm{M}$; (iv) the eigenvectors are orthogonal.

11) Find the eigenvalues and eigenvectors ($\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$), of the matrix

$$\mathrm{A} = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 3 & -2 \\ 0 & -2 & 2 \end{pmatrix}.$$

Let $\mathbf{y} = (6, -3, 0)^{\mathrm{T}}$. Observe that the vectors $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ can be used as an orthonormal basis. By taking dot products with each of the eigenvectors in turn, find the co-ordinates of $\mathbf{y}$ in this basis.

[Recall: the co-ordinates (p_1, p_2, p_3) of $\mathbf{y}$ obey $\mathbf{y} = p_1\mathbf{e}_1 + p_2\mathbf{e}_2 + p_3\mathbf{e}_3$.]

Let (q_1, q_2, q_3) be the co-ordinates of some vector $\mathbf{x}$, so $\mathbf{x} = q_1\mathbf{e}_1 + q_2\mathbf{e}_2 + q_3\mathbf{e}_3$. Use your expressions for $\mathbf{x}$ and $\mathbf{y}$ to solve the equation $\mathrm{A}\mathbf{x} = \mathbf{y}$. Is the solution unique?

12) Verify that the following matrix is orthogonal:

$$\mathrm{O} = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ -\sqrt{2} & \sqrt{2} & 0 \end{pmatrix}.$$

The linear map $f(\mathbf{x}) = \mathrm{O}\mathbf{x}$ is a rotation about some axis. If $\mathbf{x}$ is aligned with this axis, what can you say about $f(\mathbf{x})$? Explain why 1 must be an eigenvalue of O and find the corresponding axis of rotation.

Harder part: By considering geometrically the effect of the matrix on vectors that lie in the 2D subspace orthogonal to the axis, deduce that such vectors rotate according to the orthogonal matrix (defined to act in the 2D subspace)

$$\mathrm{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

where θ is the angle of rotation. By considering the characteristic polynomial of R, show that its eigenvalues are $e^{i\theta}, e^{-i\theta}$. Hence deduce that $\operatorname{trace}\mathrm{O} = 1 + 2\cos\theta$, and find the angle θ associated with the rotation matrix O.

13) For the real matrix

$$\mathrm{M} = \begin{pmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{pmatrix},$$

show with the aid of row operations that

$$\det\mathrm{M} = (a-b)(b-c)(c-a)(ab+bc+ca).$$

14) Show by direct evaluation of the determinant that

$$\Delta(x,y,z) = \begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz.$$

Now show by row or column operations that

$$x + y + z, \qquad x + \omega y + \omega^2 z, \qquad x + \omega^2 y + \omega z$$

are factors of $\Delta(x, y, z)$, where ω is a complex cube root of unity. By considering the coefficients of x^3, deduce that $\Delta(x, y, z)$ is the product of the three factors above.

15) For each real value of t, determine whether or not there exist solutions to the simultaneous equations

$$\begin{aligned} x + y + z &= t \\ tx + 2z &= 3 \\ 3x + ty + 5z &= 7 \end{aligned}$$

exhibiting the most general form of such solutions when they exist.

16) Determine the eigenvalues and eigenvectors of the symmetric matrix

$$\mathrm{A} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Use an identity of the form $\mathrm{P}^{\mathrm{T}}\mathrm{AP} = \mathrm{D}$, where D is a diagonal matrix, to find A^{-1}.

17) Find all eigenvalues, and an orthonormal set of eigenvectors, of the matrices

$$\mathrm{A} = \begin{pmatrix} 5 & 0 & \sqrt{3} \\ 0 & 3 & 0 \\ \sqrt{3} & 0 & 3 \end{pmatrix} \quad \text{and} \quad \mathrm{B} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

18) Decide whether each matrix below is an elementary matrix.

(a) $\begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 9 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

19) In each part, an elementary matrix E and a matrix A are given. Write down the row operation corresponding to E and show that the product EA results from applying the row operation to A.

(a) $\mathrm{E} = \begin{bmatrix} -6 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathrm{A} = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

(b) $\mathrm{E} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathrm{A} = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$

(c) $\mathrm{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathrm{A} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

20) Find an elementary matrix E that satisfies the equation.

(a) $\mathrm{EB} = \mathrm{D}$

(b) $\mathrm{ED} = \mathrm{B}$

(c) $\mathrm{EB} = \mathrm{F}$

(d) $\mathrm{EF} = \mathrm{B}$

$$\mathrm{A} = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, \qquad \mathrm{B} = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix},$$

$$\mathrm{C} = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}, \qquad \mathrm{D} = \begin{bmatrix} 8 & 1 & 5 \\ -6 & 21 & 3 \\ 3 & 4 & 1 \end{bmatrix},$$

$$\mathrm{F} = \begin{bmatrix} 8 & 1 & 5 \\ 8 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

21) Invert the following matrices, if it is possible:

(a) $\begin{bmatrix} -1 & 3 \\ 3 & -2 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$

(d) $\begin{bmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{10} \\ \frac{1}{5} & -\frac{4}{5} & \frac{1}{10} \end{bmatrix}$

(e) $\begin{bmatrix} -8 & 17 & 2 & \frac{1}{3} \\ 4 & 0 & \frac{2}{5} & -9 \\ 0 & 0 & 0 & 0 \\ -1 & 13 & 4 & 2 \end{bmatrix}$

(f) $\begin{bmatrix} -1 & 0 & 1 & 0 \\ 2 & 3 & -2 & 6 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 1 & 5 \end{bmatrix}$

22) For what values of c are the following matrices invertible?

(a) $\begin{bmatrix} c & c & c \\ 1 & c & c \\ 1 & 1 & c \end{bmatrix}$

(b) $\begin{bmatrix} c & 1 & 0 \\ 1 & c & 1 \\ 0 & 1 & c \end{bmatrix}$

23) Given the 3×3 matrix:

$$\mathrm{A} = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 1 \\ -6 & -1 & 2 \end{bmatrix}$$

Find the LU decomposition of A.

24) Find the LU decomposition of

$$A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}$$